The Schematics have been analyzed from
a die picture by InfosecDJ.
The chip is identified as:

MSM OKi
79H041

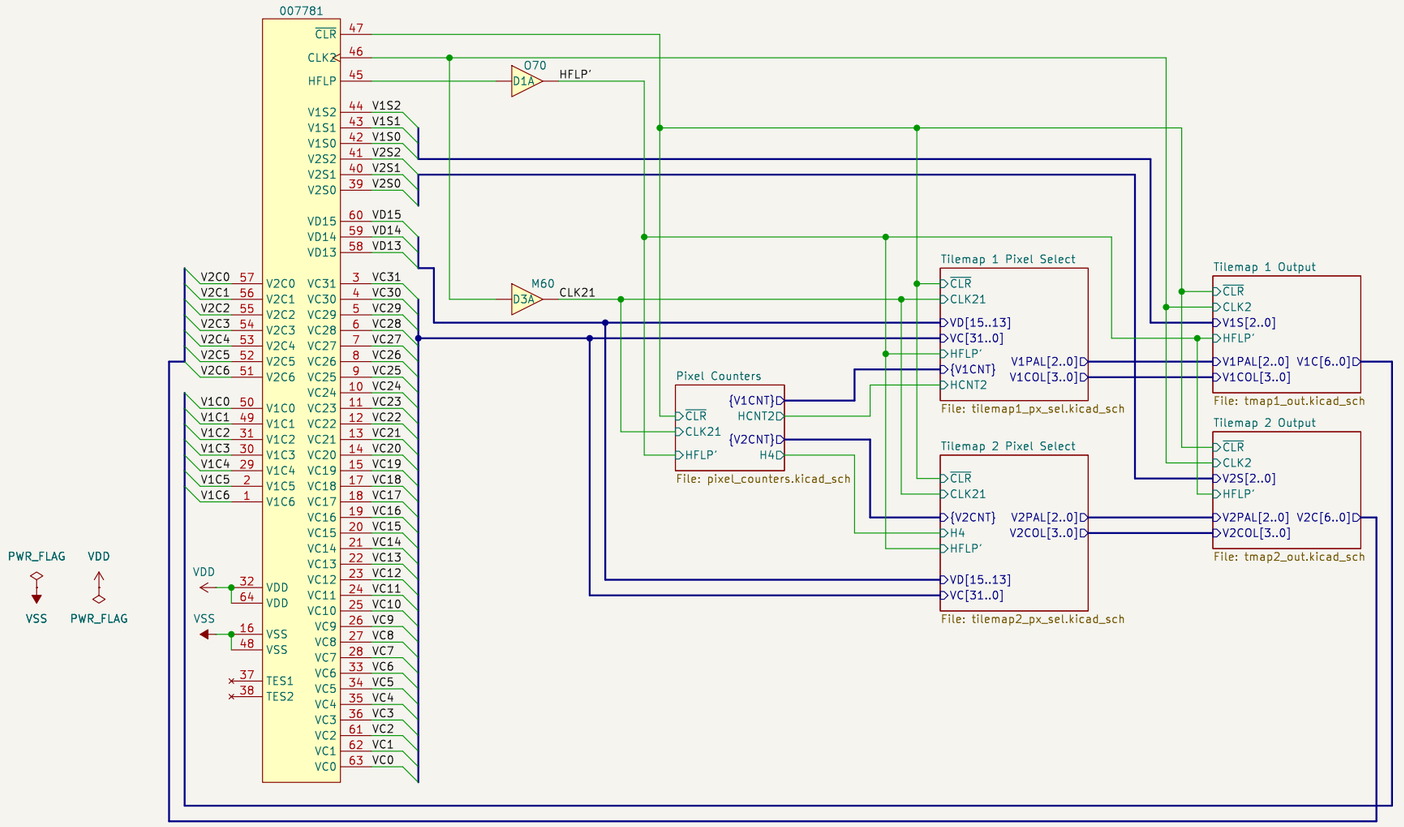
It is a MSM79V000 gate array
from the MSM70V000 series.
– 3289 Basic Cells
– 23 Columns, 143 Rows

The konami 007781 is a tilemap pixel shift register
with scroll. The tiles are 8x8 pixels, but the 007781
don't care about vertical line position. 007781 handles
two tilemaps at a time:
– V1, tilemap 1
– V2, tilemap 2

8 pixels, 4 bits each, of color data at a time are latched
into the registers. The palette code is latched 8 pixel
clocks earlier from VRAM. This is because the
data in the 007780 DRAM controller and DRAM takes 8
pixel clocks to process.

V1 latches pixels when the horizontal counter, going
from 0 to 7, goes to 2 (HCNT2) and V2 latches when
the horizontal counter goes to 4.
Data latched for:
– V1, when HCNT[2:0] == 2
– V2, when HCNT[2:0] == 4

The pixel code and palette color data are shifted
out between 1 to 9 pixel clocks after HCNT[2:0] == 4,
depending on if scroll or horizontal flip are active.

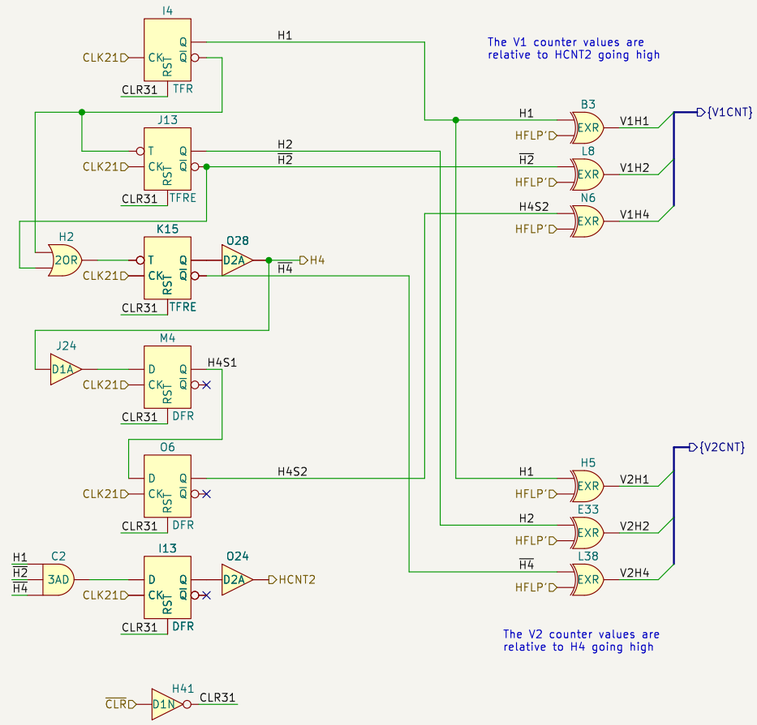


Ulf Skutnabba, twitter: @skutis77

Sheet: /
File: 007781.kicad_sch

Title: Konami 007781

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/6



Sheet: /Pixel Counters/		
File: pixel_counters.kicad_sch		
Title: Konami 007781		
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 2/6

V1PAL[2..0]
V1PAL2
V1PAL1
V1PAL0
V1COL[3..0]
V1COL3
V1COL2
V1COL1
V1COL0
CLR
CLK2
D1N
D1A
DFR
Q70
T77
C63
K84
M70
U100
M95
M111
J93
N107
R65
L81
K69
R82
V1PAL2[1..8]
V1PAL21 V1PAL22 V1PAL23 V1PAL24 V1PAL25 V1PAL26 V1PAL27 V1PAL28
O74 F78 H77 J83 O84 N82 Q80 P70
V1PAL1[1..8]
V1PAL11 V1PAL12 V1PAL13 V1PAL14 V1PAL15 V1PAL16 V1PAL17 V1PAL18
B63 E83 A89 B85 B74 C68 C78 D84
V1PAL0[1..8]
V1PAL01 V1PAL02 V1PAL03 V1PAL04 V1PAL05 V1PAL06 V1PAL07 V1PAL08
D69 R85 O94 R94 W111 Q89 U90 V96
V1COL3[1..8]
V1COL31 V1COL32 V1COL33 V1COL34 V1COL35 V1COL36 V1COL37 V1COL38
N98 H124 N127 I124 Q124 P124 P134 Q134
V1COL2[1..8]
V1COL21 V1COL22 V1COL23 V1COL24 V1COL25 V1COL26 V1COL27 V1COL28
J102 G134 H134 H114 H104 F104 L99 M99
V1COL1[1..8]
V1COL11 V1COL12 V1COL13 V1COL14 V1COL15 V1COL16 V1COL17 V1COL18
U58 N73 V85 W96 W85 V72 R69 U68
V1COL0[1..8]
V1COL01 V1COL02 V1COL03 V1COL04 V1COL05 V1COL06 V1COL07 V1COL08
F55 E73 A72 F65 G65 H67 J73 L73
P79 M80 M33 V1C6
E93 F88 G75 V1C5
T83 S90 T80 V1C4
M125 L116 K124 V1C3
K100 I95 G104 V1C2
T60 S73 W80 V1C1
K73 I67 L70 V1C0
V1C[6..0]
V1SF[2..0]
V1S[2..0]
V1S2 V1S1 V1S0
HFLP'D
P64 P90 R78
EXR
V1SF2 V1SF1 V1SF0
INV1
T73B T73A T75B T75A T71B T71A
I84B I84A U84B U84A G79B G79A
K87B K87A U87B U87A U81B U81A
T94B T94A V107B V107A K98B K98A
W107B W107A D80B D80A I78B I78A
V82B V82A U78B U78A P68B P68A
F75B F75A Q78B Q78A M74B M74A
Scroll
Sheet: /Tilemap 1 Output/
File: tmap1_out.kicad_sch
Title: Konami 007781
Size: A3
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V2PAL[2..0]
V2PAL2
V2PAL1
V2PAL0
V2COL[3..0]
V2COL3
V2COL2
V2COL1
V2COL0
V2PAL2[1..8]
V2PAL1[1..8]
V2PAL0[1..8]
V2COL3[1..8]
V2COL2[1..8]
V2COL1[1..8]
V2COL0[1..8]
V2C[6..0]
V2C6
V2C5
V2C4
V2C3
V2C2
V2C1
V2C0
V2S[2..0]
V2SF[2..0]
Scroll
Sheet: /Tilemap 2 Output/
File: tmap2_out.kicad_sch
Title: Konami 007781
Size: A3
Date: 2024-09-08
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The pixel data is latched when
HCNT2 goes high.
The pixel data order is

1. VC31,VC30,VC29,VC28
2. VC27,VC26,VC25,VC24
3. VC23,VC22,VC21,VC20
4. VC19,VC18,VC17,VC16
5. VC15,VC14,VC13,VC12
6. VC11,VC10,VC9,VC8
7. VC7,VC6,VC5,VC4
9. VC3,VC2,VC1,VC0

The palette color is delayed 8 pixels. It is because
the 007780 takes 8 pixels to process the pixel data.

Sheet: /Tilemap 1 Pixel Select/	
File: tilemap1_px_sel.kicad_sch	
Title: Konami 007781	
Size: A3	Date: 2024-09-08
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VC[31..0]

The pixel data is latched when
HC4 goes high.
The pixel data order is

1. VC31,VC30,VC29,VC28
2. VC27,VC26,VC25,VC24
3. VC23,VC22,VC21,VC20
4. VC19,VC18,VC17,VC16
5. VC15,VC14,VC13,VC12
6. VC11,VC10,VC9,VC8
7. VC7,VC6,VC5,VC4
8. VC3,VC2,VC1,VC0

{V2CNT}

When flip is not active
pixel data is delayed one pixel.

VD[15..13]

V2COL[3..0]

V2PAL[2..0]

Sheet: /Tilemap 2 Pixel Select/
File: tilemap2_px_sel.kicad_sch

Title: Konami 007781

Size: A3 | Date: 2024-09-08 | Rev:
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